

By Theorem 3, the Matrix equation has the same solution set as the system of linear equations corresponding to the augmented matrix $[v_1 \ v_2 \ \dots \ v_p \ 0]$. Since the system is consistent (trivial solution), we claim that there exists free variables. The augmented matrix, on application of the Row Reduction algorithm and Theorem 1 can be converted into a unique rref.

- The following conditions, according to defn. of rref are True:
 - For each row, the leading element (which must be 1) is in a column to the right of the columns of the leading elements of all rows above it.
 - There can be at most n columns having leading elements ($\therefore n$ rows are present) and at least $(p-n)$ free variables are there.

By Theorem 2, the system of equations has infinitely many solns.

$\therefore$ There exist solutions where not all of the weights are zero but $c_1 v_1 + c_2 v_2 + \dots + c_p v_p = 0$ $\therefore \{v_1, v_2, \dots, v_p\}$ are linearly dependent $\square$

Warning: Theorem 8 says nothing about the case in which the number of vectors in the set does not exceed the number of entries in each vector.

Example 5: The vectors $\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \end{bmatrix}$ are linearly dependent by Theorem 8, because there are three vectors in the set and there are only 2 entries in each vector. $\square$

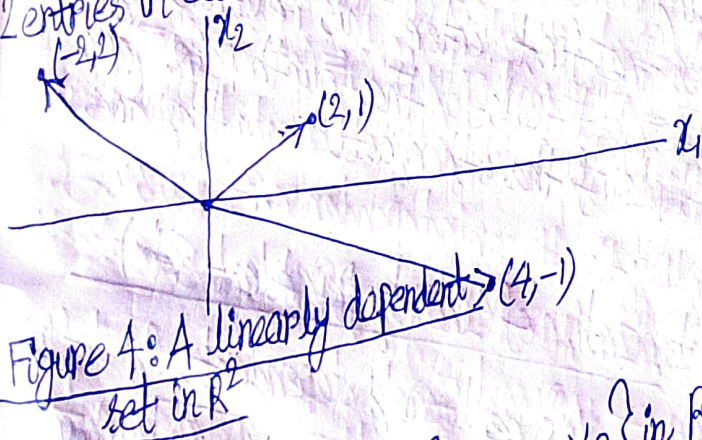


Figure 4: A linearly dependent set in $\mathbb{R}^2$

Theorem 9: If a set $S = \{v_1, \dots, v_p\}$ in $\mathbb{R}^n$ contains the zero vector, then the set is linearly dependent.